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ORF307: Optimization

Spring 2025 1$^{\text{st}}$ Midterm Exam Solutions

March 6, 11:00am – 12:20pm

Exam Rules

1. Please write out and sign the following pledge on the solution booklet: "I pledge my honor that I have not violated the Honor Code or the rules specified by the instructor during this exam."
2. You are allowed a single sheet of paper, double-sided, hand-written or typed.
3. You cannot communicate with anyone during the exam.
4. All questions should be answered in the solution booklet. Write your name on every page that you use to write the solutions. Start each problem on a separate page. Answering multiple subparts of the same problem on the same page is fine. We will **not** grade anything written on the exam questions; we will only grade what you write in the solution booklet.
5. There are 3 problems worth a total of 100 points.
6. At the end of the exam, please turn in both your solution booklet and the exam questions.

1. *Problem 1 (30 points). Portfolio Dilemmas: Tech vs Coffee?*

   You are deciding how to allocate your investments between two assets: a **tech stock** and a **coffee chain stock**. The tech stock has historically offered higher returns but is more volatile, while the coffee chain stock is more stable but has lower returns. You want to construct a portfolio that balances return and risk using historical data.

   Let us first consider a more general setting in which we have past return data for T trading days across n assets, stored in the return matrix $R \in \mathbf{R}^{T\times n}$, where R_{ti} is the return of asset i on day t. The vector of *average returns per asset* is

   $$\mu = \frac{1}{T}R^T\mathbf{1} \in \mathbf{R}^n.$$

   Our goal is to find a portfolio allocation $w = (w_1, \ldots, w_n)$ over the n assets to achieve a target portfolio return $\rho = \mu^T w$ while minimizing risk. The *portfolio return time series* is given by

   $$r = Rw,$$

   and the *risk* is defined as

   $$\mathbf{std}(r) = \|Rw - \rho\mathbf{1}\|_2.$$

   a) (8p) Formulate the portfolio optimization problem as a constrained least squares problem that minimizes the squared risk while ensuring that the expected portfolio return meets the target ρ. Write down the KKT conditions for this problem.

   Let us now go back to our original problem and assume that you have **only two stocks**, *i.e.*, $n = 2$, with $\mu_1 \neq \mu_2$ representing the coffee chain and the tech stock, respectively.

   b) (12p) Find the optimal allocations w_1 and w_2.
   Hint: The optimal weights w depend only on μ and ρ and not directly on R.

   c) (10p) Assume $\mu_1 < \mu_2$, meaning the tech stock has a higher average return. Characterize for which values of ρ the optimal portfolio will:

      – Hold *long positions in both stocks*,
      – Hold *one long and one short position*,
      – Hold *short positions in both stocks*.

      Hint: The answer should depend on how the target return ρ compares to μ_1 and μ_2.

2. *(35 points). Noisy Measurements: Estimating the True Ratings*

   You are trying to estimate the true ratings of four Princeton restaurants, labeled as $x = (x_1, x_2, x_3, x_4) \in \mathbf{R}^4$. However, you know that some restaurants are managed by the same company, which enforces consistency in their ratings:

$$x_1 - 0.8x_2 = 0, \quad x_3 - 0.8x_4 = 0.$$

   To estimate these ratings, you collect user-submitted scores, but the data is noisy. Specifically, you have access to approximate pairwise comparisons, meaning that for each pair of restaurants, you observe:

$$y_{ij} \approx x_i + x_j, \quad 1 \leq i < j \leq 4,$$

   where $\{y_{ij}\}_{1 \leq i < j \leq 4}$ are known measurements based on user reviews.

   To determine the best estimate for the true ratings, you decide to minimize the *sum of squares of the residuals* $x_i + x_j - y_{ij}$ over all such pairs.

   (a) (15p) Show that this problem can be written as a constrained least squares problem of the form

$$\begin{array}{ll} \text{minimize} & \|Ax - b\|^2 \\ \text{subject to} & Cx = d. \end{array}$$

   for appropriate matrices A, C and vectors b, d. Write down the dimensions of A, b, C, d and their entries.

   (b) (15p) Write the *KKT linear system* for your formulation in part (a). Your final answer should be in terms of $\{y_{ij}\}_{1 \leq i < j \leq 4}$ and the solution vector.

   (c) (5p) For any noisy measurements $\{y_{ij}\}_{1 \leq i < j \leq 4}$, is the solution to your formulation always unique? Explain why or why not.

3. *(35 points). Optimal Trade Execution: The Art of Not Moving the Market.*

As a trader for a hedge fund, your mission is to buy M dollars worth of a hot stock over a fixed period. But there's a catch: if you buy too aggressively, the market might catch on, driving prices up before you complete your purchase. On the other hand, if you buy too slowly, you risk missing a good price window.

The stock price at time t is given by $p_t \in \mathbf{R}$, and market liquidity constraints prevent you from purchasing more than L dollars of shares per unit time. We denote the full price trajectory as $p = (p_1, \ldots, p_T) \in \mathbf{R}^T$.

Your job is to determine the optimal purchasing strategy $x = (x_1, \ldots, x_T) \in \mathbf{R}^T$, where x_t represents the dollar value of shares bought at time t.

(a) (10p) Suppose you have insider-level foresight and know the full price trajectory p in advance. Formulate a linear program to minimize the total cost of acquiring the required shares while ensuring that purchases remain non-negative and within the liquidity constraints.

(b) (15p) To avoid sending shockwaves through the market, you add a penalty discouraging sudden large trades. Specifically, you minimize the total cost from (a) while also discouraging fluctuations in purchases between consecutive time steps. The penalty term is given by:

$$\lambda \sum_{t=1}^{T-1} |x_{t+1} - x_t|,$$

where $\lambda > 0$ controls how smooth the purchases should be. Formulate the corresponding linear program.

(c) (10p) Of course, in reality, no one has a crystal ball. Instead, you have access to K possible price trajectories $p^1, p^2, \ldots, p^K \in \mathbf{R}^T$. To hedge against the worst-case scenario, formulate a linear program that minimizes the highest (worst-case) possible cost across all potential price paths.
Hint: The worst-case cost does not affect the penalty introduced in part (b).